

AMON KIPNGETICH KIPROP

Senior AI Engineer & Machine Learning Specialist | Swahili Linguistics Expert

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PROFESSIONAL SUMMARY

Results-driven Senior AI Engineer and Machine Learning Specialist with over 10 years of experience designing, developing, and deploying scalable enterprise software solutions. Holds advanced degrees in Computer Science (AI/ML focus) and Swahili Linguistics, combining deep technical expertise with rare linguistic and cultural insight into East African contexts. Proven track record leading cross-functional teams, building data-driven products, and delivering high-impact digital transformation projects across fintech, healthcare, and government sectors in Kenya and internationally. Founder of Amon.Academy, an AI and data science educational platform serving learners across East Africa.

EDUCATION

Master of Science in Computer Science (AI/ML Specialisation) | University of Nairobi | Graduated: 2016

Thesis: Distributed Machine Learning Architectures for Resource-Constrained Environments

Bachelor of Arts in Swahili Linguistics | University of Nairobi | Graduated: 2014

Focus areas: Applied Linguistics, Language Technology, NLP for Swahili, Sociolinguistics of East African Languages | First Class Honours

Bachelor of Science in Computer Science | JKUAT, Nairobi | Graduated: 2013

First Class Honours

PROFESSIONAL EXPERIENCE

Senior Software Engineer & Team Lead | Safaricom PLC | January 2021 – Present

- Lead a team of 8 engineers building and maintaining M-Pesa's next-generation API gateway processing 5 million+ daily transactions.
- Architected and deployed a cloud-native microservices platform on AWS, reducing system downtime by 40% and saving KES 12M annually in infrastructure costs.
- Spearheaded DevOps transformation including CI/CD pipelines, Kubernetes container orchestration, and automated testing frameworks.
- Built and deployed a real-time fraud detection ML model (TensorFlow) achieving 96.4% accuracy, reducing fraudulent transaction losses by 28%.
- Led data pipeline modernisation using Apache Kafka and Airflow, enabling near-real-time analytics dashboards for business intelligence.
- Collaborated with product managers and stakeholders to define technical roadmaps and deliver quarterly milestones on schedule.

Software Engineer II | Andela Kenya | March 2018 – December 2020

- Developed full-stack web applications for international clients using React.js, Node.js, and PostgreSQL, serving user bases of 50,000+.
- Designed and integrated NLP-based text classification pipelines (Scikit-learn, spaCy) for a U.S. legal tech client, reducing document review time by 45%.
- Built RESTful and GraphQL APIs for a U.S. healthcare SaaS platform with HIPAA-compliant data storage and processing.
- Improved application performance by 35% through PostgreSQL query optimisation, indexing strategies, and Redis caching.
- Mentored 15 junior developers through structured learning paths, code reviews, and pair programming sessions.
- Integrated ETL pipelines for client reporting systems using Python, Pandas, and Redshift.

Software Developer | *iHub Nairobi / Consulting* | June 2016 – February 2018

- Designed and delivered custom software solutions for 10+ startups and SMEs across East Africa, including ERP systems, mobile apps, and analytics dashboards.
- Developed a livestock management mobile app for a Kenyan agri-tech startup, integrating a crop demand forecasting model (XGBoost) — reached 8,000 active farmers within 6 months.
- Built NLP-powered citizen feedback classifier for a Government of Kenya pilot, processing and categorising 20,000+ survey responses automatically.
- Integrated M-Pesa Daraja API and Paystack payment gateways across multiple e-commerce platforms.
- Conducted technical workshops in Python, machine learning fundamentals, and database design for 200+ participants.

Junior Software Developer | *Kenya ICT Authority* | August 2013 – May 2016

- Contributed to government e-service portals used by over 2 million Kenyan citizens, including Huduma Centre digital services.
- Worked on data digitisation projects migrating legacy paper records to secure cloud-based document management systems.
- Built and maintained ETL workflows to consolidate data from 14 government ministries onto a unified service platform.
- Performed database administration, query tuning, and data quality audits on Oracle and MySQL databases.
- Wrote technical documentation, data dictionaries, and user guides for internal IT systems and public-facing services.

SWAHILI LINGUISTICS EXPERTISE

Combining a formal degree in Swahili Linguistics with AI/ML specialisation, I bridge language technology and East African linguistic contexts — a rare and highly valuable skill set for AI training, NLP research, and localisation projects.

Core Competencies	Morphology & syntax analysis, phonology, semantics, pragmatics, discourse analysis, sociolinguistics
Language Technology	Swahili NLP pipeline development, tokenisation, POS tagging, named entity recognition, machine translation
AI Data Tasks	Native-level Swahili text annotation, AI response evaluation, cultural appropriateness review, pairwise comparison, data labelling
Dialects & Variants	Standard Swahili (Kiswahili Sanifu), Kenyan Swahili (Sheng influence), Coastal Swahili, Tanzanian Swahili
Content Expertise	Translation & localisation, transcription, subtitling, quality review, culturally sensitive content moderation
Research Areas	Corpus linguistics for Swahili, Bantu language family, language endangerment and preservation, code-switching in urban Kenya

TECHNICAL SKILLS

Programming	Python, JavaScript (ES6+), TypeScript, Java, R, SQL, Bash/Shell
Frameworks	React.js, Node.js, Express, Django, FastAPI, Spring Boot, GraphQL
Cloud & DevOps	AWS (EC2, S3, Lambda, SageMaker, RDS), GCP, Docker, Kubernetes, GitHub Actions, Jenkins, Terraform
Databases	PostgreSQL, MySQL, MongoDB, Redis, Firebase, Cassandra, Snowflake
Tools	Git, Jira, Postman, VS Code, IntelliJ IDEA, Jupyter Notebook, Apache Kafka, Apache Spark
Methodologies	Agile/Scrum, TDD, CI/CD, Microservices, REST, SOLID Principles, Domain-Driven Design

AI & MACHINE LEARNING

ML Frameworks	TensorFlow, PyTorch, Scikit-learn, Keras, Hugging Face Transformers, XGBoost, LightGBM
AI Specialties	NLP, Computer Vision, Predictive Analytics, Recommendation Systems, Time-Series Forecasting
LLMs & GenAI	Prompt Engineering, RAG (Retrieval-Augmented Generation), Fine-tuning LLMs, OpenAI API, LangChain, Vector Databases (Pinecone, Weaviate)
MLOps	MLflow, Weights & Biases, AWS SageMaker, model versioning, A/B testing, automated monitoring & retraining pipelines
Key AI Projects	Fraud detection model 96.4% accuracy (M-Pesa); NLP citizen feedback classifier (GoK); crop demand forecasting engine (agri-tech)

DATA MANAGEMENT & ANALYTICS

Data Engineering	ETL/ELT pipeline design, Apache Airflow, Apache Spark, Hadoop, dbt, Kafka Streams
Data Warehousing	Snowflake, Google BigQuery, Amazon Redshift, data lake architecture on AWS S3
BI & Visualisation	Tableau, Power BI, Looker, Metabase, Matplotlib, Seaborn, Plotly
Data Governance	Data quality frameworks, GDPR & Kenya Data Protection Act compliance, data lineage, MDM

LANGUAGE PROFICIENCY

Language	Proficiency Level	Notes
Swahili (Kiswahili)	Native / Bilingual (C2)	Formal degree in Swahili Linguistics; expert in written, spoken, and literary Swahili
English	Full Professional (C2)	Advanced reading, writing, and spoken communication
French	Professional Working (B2)	Reading, writing, and conversational proficiency
Kalenjin	Native Speaker	Mother tongue; fluency in spoken Kalenjin

EDUCATIONAL LEADERSHIP

Founder & Lead Instructor | *Amon.Academy* | 2022 – Present

- Founded Amon.Academy, an AI and data science educational platform delivering structured courses across Python, machine learning, NLP, and data engineering for East African learners.
- TSC-certified educator with experience designing and delivering curriculum for diverse audiences from university students to working professionals.
- Developed interactive learning tools and platforms (PyOOP Academy, Pandas Academy, DataCore Academy, PyLingo) to make AI education accessible and engaging.
- Conducted technical workshops on Python, ML fundamentals, and database design for 200+ participants across Nairobi's tech ecosystem.

CERTIFICATIONS & AWARDS

- AWS Certified Solutions Architect – Associate (2022)
- Google Professional Cloud Developer Certificate (2021)
- TensorFlow Developer Certificate – Google (2021)
- Meta Front-End Developer Professional Certificate (2020)
- TSC-Certified Educator – Teachers Service Commission, Kenya

- Safaricom PLC Innovation Award – Best Engineering Team (2022)

REFERENCES

Available upon request.